

ASSIGNMENT 1

Q1 -----topo1

ii.

mininet> h1 traceroute h2

traceroute to 10.0.0.2 (10.0.0.2), 30 hops max, 60 byte packets

1 10.0.0.2 (10.0.0.2) 0.809 ms 0.964 ms 0.939 ms

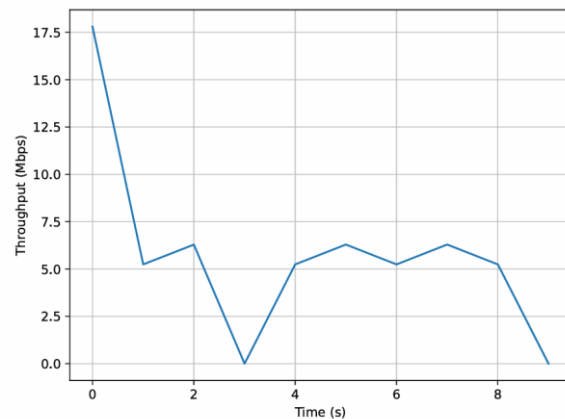
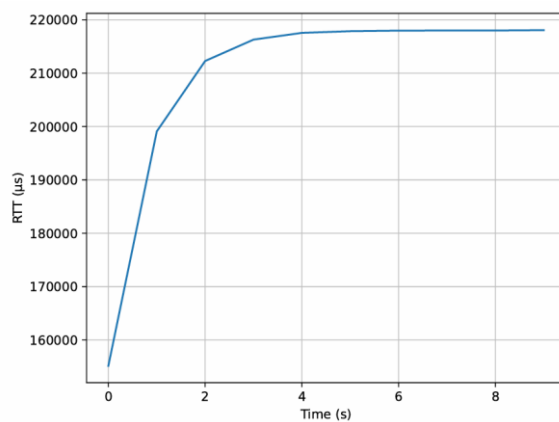
iii. throughput calculation

a. Time=10sec interval=1 , bw = 5mbps

```

root@vaishnavi:/home/vaishnavi# iperf3 -c 10.0.0.2 -t 10 -i 1
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 56766 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate    Retr    Cwnd
[ 7] 0.00-1.00    sec 2.12 MBytes 17.8 Mbits/sec 0    383 KBytes
[ 7] 1.00-2.00    sec 768 KBytes 6.29 Mbits/sec 0    383 KBytes
[ 7] 2.00-3.00    sec 896 KBytes 7.34 Mbits/sec 0    383 KBytes
[ 7] 3.00-4.00    sec 0.00 Bytes 0.00 bits/sec 0    383 KBytes
[ 7] 4.00-5.00    sec 768 KBytes 6.29 Mbits/sec 0    383 KBytes
[ 7] 5.00-6.00    sec 896 KBytes 7.34 Mbits/sec 0    383 KBytes
[ 7] 6.00-7.00    sec 768 KBytes 6.29 Mbits/sec 0    383 KBytes
[ 7] 7.00-8.00    sec 0.00 Bytes 0.00 bits/sec 0    383 KBytes
[ 7] 8.00-9.00    sec 896 KBytes 7.34 Mbits/sec 0    383 KBytes
[ 7] 9.00-10.00   sec 768 KBytes 6.28 Mbits/sec 0    383 KBytes
-----
[ ID] Interval      Transfer    Bitrate    Retr    sender
[ 7] 0.00-10.00   sec 7.75 MBytes 6.50 Mbits/sec 0
[ 7] 0.00-10.22   sec 5.75 MBytes 4.72 Mbits/sec
receiver

```

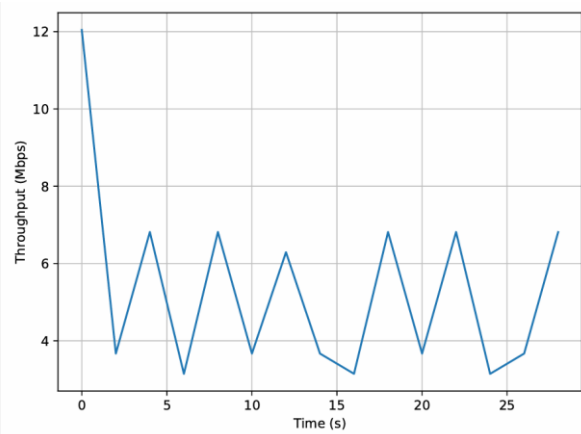
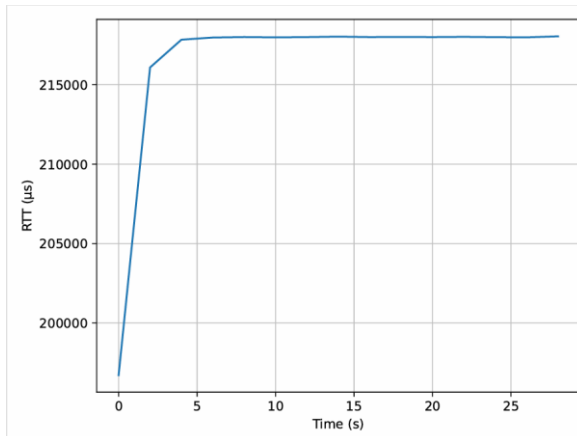


b. time=30sec interval=2

```

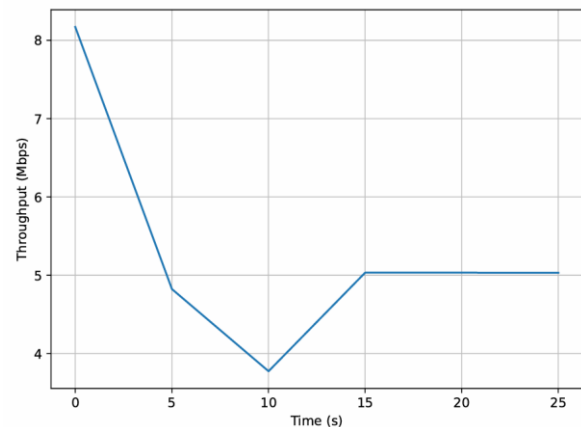
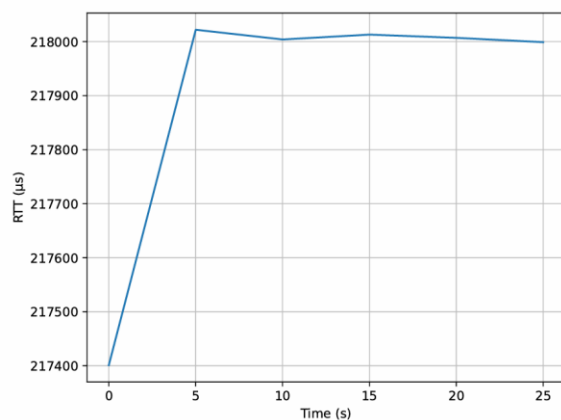
[ 7] local 10.0.0.1 port 57916 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate    Retr    Cwnd
[ 7] 0.00-2.00    sec 2.62 MBytes 11.0 Mbits/sec 0    356 KBytes
[ 7] 2.00-4.00    sec 1.50 MBytes 6.29 Mbits/sec 0    356 KBytes
[ 7] 4.00-6.00    sec 768 KBytes 3.15 Mbits/sec 0    356 KBytes
[ 7] 6.00-8.00    sec 1.38 MBytes 5.77 Mbits/sec 0    356 KBytes
[ 7] 8.00-10.00   sec 1.50 MBytes 6.29 Mbits/sec 0    356 KBytes
[ 7] 10.00-12.00   sec 768 KBytes 3.15 Mbits/sec 0    356 KBytes
[ 7] 12.00-14.00   sec 1.50 MBytes 6.29 Mbits/sec 0    356 KBytes
[ 7] 14.00-16.00   sec 768 KBytes 3.15 Mbits/sec 0    356 KBytes
[ 7] 16.00-18.00   sec 1.50 MBytes 6.29 Mbits/sec 0    356 KBytes
[ 7] 18.00-20.00   sec 768 KBytes 3.15 Mbits/sec 0    356 KBytes
[ 7] 20.00-22.00   sec 1.50 MBytes 6.29 Mbits/sec 0    356 KBytes
[ 7] 22.00-24.00   sec 768 KBytes 3.15 Mbits/sec 0    356 KBytes
[ 7] 24.00-26.00   sec 1.50 MBytes 6.29 Mbits/sec 0    356 KBytes
[ 7] 26.00-28.00   sec 768 KBytes 3.15 Mbits/sec 0    356 KBytes
[ 7] 28.00-30.00   sec 1.50 MBytes 6.29 Mbits/sec 0    356 KBytes
-----
[ ID] Interval      Transfer    Bitrate    Retr    sender
[ 7] 0.00-30.00   sec 13.0 MBytes 5.31 Mbits/sec 0
[ 7] 0.00-30.26   sec 17.1 MBytes 4.75 Mbits/sec
receiver

```



c. time=30sec interval=5

```
iperf Done.
root@vaishnavi:/home/vaishnavi# iperf3 -c 10.0.0.2 -t 30 -i 5
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 43066 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate    Retr  Cwnd
[ 7] 0.00-5.01 sec  4.50 MBytes  7.54 Mbits/sec  0    383 KBytes
[ 7] 5.01-10.01 sec  3.25 MBytes  5.45 Mbits/sec  0    383 KBytes
[ 7] 10.01-15.01 sec  2.38 MBytes  3.98 Mbits/sec  0    383 KBytes
[ 7] 15.01-20.01 sec  3.25 MBytes  5.45 Mbits/sec  0    383 KBytes
[ 7] 20.01-25.01 sec  2.50 MBytes  4.19 Mbits/sec  0    383 KBytes
[ 7] 25.01-30.01 sec  3.25 MBytes  5.45 Mbits/sec  0    383 KBytes
-----
[ ID] Interval      Transfer    Bitrate    Retr
[ 7] 0.00-30.01 sec  19.1 MBytes  5.35 Mbits/sec  0
[ 7] 0.00-30.28 sec  17.2 MBytes  4.78 Mbits/sec
iperf Done.
```



Topo2 : Code:

```
#!/usr/bin/python3
```

```
from mininet.topo import Topo
```

```
from mininet.net import Mininet
```

```
from mininet.link import TCLink
```

```
from mininet.node import OVSSwitch
```

```
from mininet.cli import CLI
```

```
from mininet.log import setLogLevel
```

```
class Topology2(Topo):
```

```
    """
```

Topology-2:

h1 --(10Mbps)-- s1 --(10Mbps)-- s2 --(10Mbps)-- h2

"""

```
def build(self):
```

```
    h1 = self.addHost('h1')
```

```
    h2 = self.addHost('h2')
```

```
    s1 = self.addSwitch('s1', failMode='standalone')
```

```
    s2 = self.addSwitch('s2', failMode='standalone')
```

```
    self.addLink(h1, s1, cls=TCLink, bw=10)
```

```
    self.addLink(s1, s2, cls=TCLink, bw=10)
```

```
    self.addLink(s2, h2, cls=TCLink, bw=10)
```

```
if __name__ == "__main__":
```

```
    setLogLevel('info')
```

```
    topo = Topology2()
```

```
    net = Mininet(
```

```
        topo=topo,
```

```
        controller=None,
```

```
        switch=OVSSwitch,
```

```
        link=TCLink,
```

```
        autoSetMacs=True,
```

```
        autoStaticArp=True
```

```
)
```

```
    net.start()
```

```
    for sw in net.switches:
```

```
        sw.cmd(f'ovs-ofctl del-flows {sw.name}')
```

```
        sw.cmd(f'ovs-ofctl add-flow {sw.name} "priority=0,actions=NORMAL"')
```

```
    print("\n*** Running pingall (should be 0% dropped now)")
```

```
    net.pingAll()
```

```
    print("\n*** Entering Mininet CLI")
```

```
    CLI(net)
```

```
    net.stop()
```

ii.

```
mininet> h1 traceroute -n h2
```

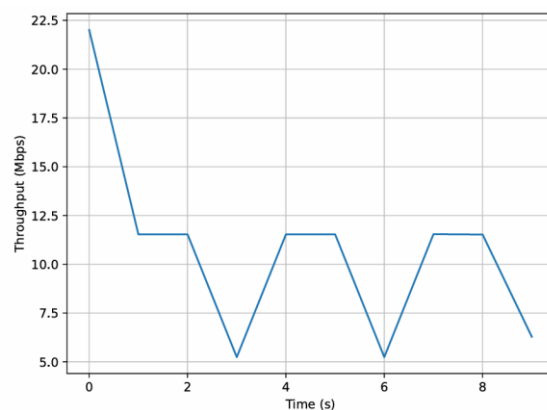
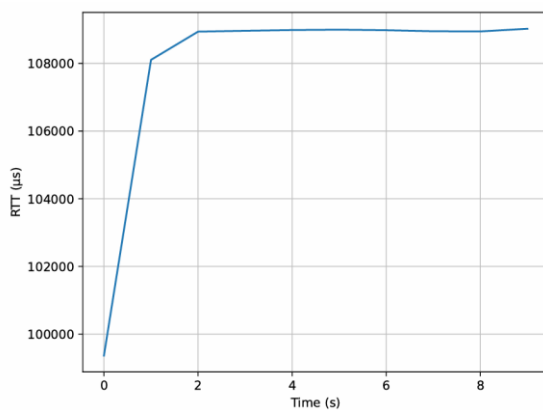
traceroute to 10.0.0.2 (10.0.0.2), 30 hops max, 60 byte packets

1 10.0.0.1 3072.820 ms !H 3072.723 ms !H 3072.057 ms !H

iii. throughput calculation

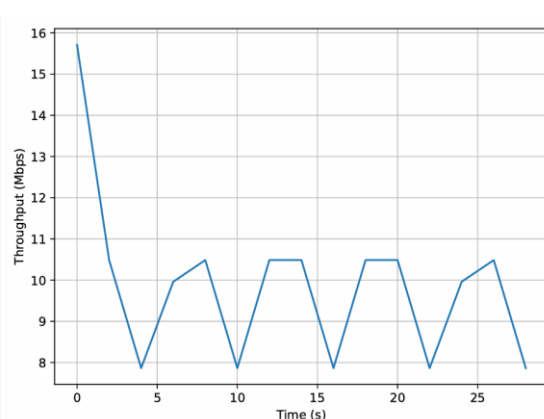
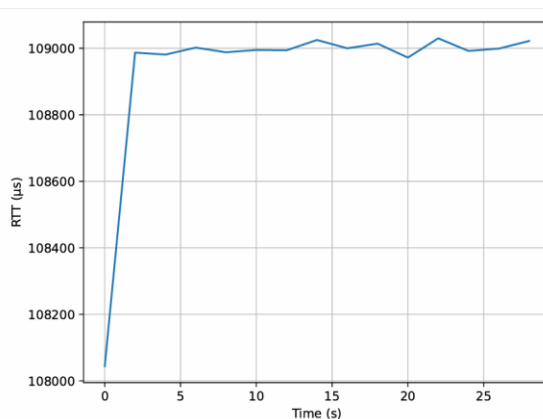
a. time=10sec interval=1

```
root@vaishnavi:/home/vaishnavi# iperf3 -c 10.0.0.2 -t 10 -i 1
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 55538 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate    Retr  Cwnd
[ 7] 0.00-1.00    sec  2.62 MBytes  22.0 Mbits/sec  0    341 KBytes
[ 7] 1.00-2.00    sec  1.38 MBytes  11.5 Mbits/sec  0    341 KBytes
[ 7] 2.00-3.00    sec  1.38 MBytes  11.5 Mbits/sec  0    341 KBytes
[ 7] 3.00-4.00    sec  640 KBytes  5.24 Mbits/sec  0    341 KBytes
[ 7] 4.00-5.00    sec  1.38 MBytes  11.5 Mbits/sec  0    341 KBytes
[ 7] 5.00-6.00    sec  1.38 MBytes  11.5 Mbits/sec  0    341 KBytes
[ 7] 6.00-7.00    sec  640 KBytes  5.24 Mbits/sec  0    341 KBytes
[ 7] 7.00-8.00    sec  1.38 MBytes  11.5 Mbits/sec  0    341 KBytes
[ 7] 8.00-9.00    sec  1.38 MBytes  11.5 Mbits/sec  0    341 KBytes
[ 7] 9.00-10.00   sec  768 KBytes  6.29 Mbits/sec  0    341 KBytes
--
[ ID] Interval      Transfer    Bitrate    Retr  sender receiver
[ 7] 0.00-10.00    sec  12.9 MBytes  10.8 Mbits/sec  0
[ 7] 0.00-10.13   sec  11.5 MBytes  9.53 Mbits/sec
```



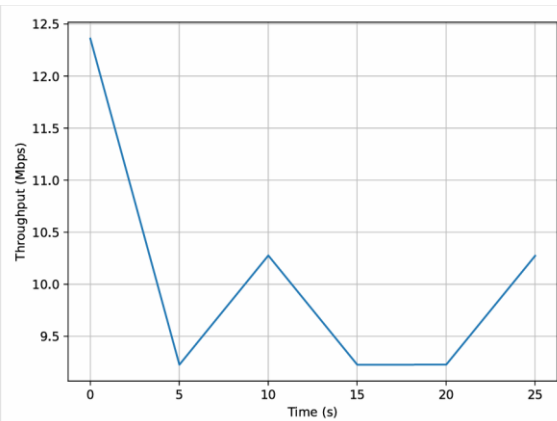
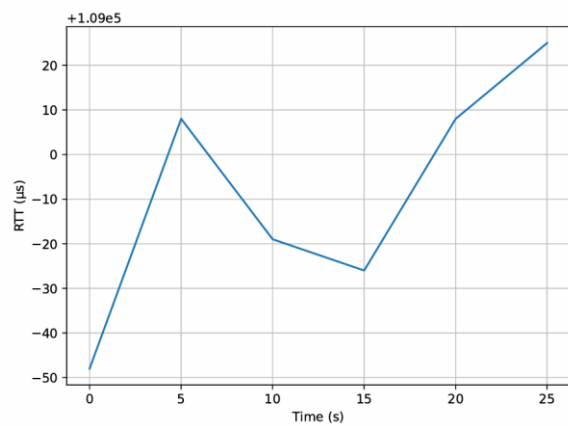
b. time=30sec interval=2

```
[ 7] local 10.0.0.1 port 44352 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate    Retr  Cwnd
[ 7] 0.00-2.00    sec  3.75 MBytes  15.7 Mbits/sec  0    383 KBytes
[ 7] 2.00-4.00    sec  2.38 MBytes  9.96 Mbits/sec  0    383 KBytes
[ 7] 4.00-6.00    sec  2.50 MBytes  10.5 Mbits/sec  0    383 KBytes
[ 7] 6.00-8.00    sec  2.38 MBytes  9.96 Mbits/sec  0    383 KBytes
[ 7] 8.00-10.00   sec  2.38 MBytes  9.96 Mbits/sec  0    383 KBytes
[ 7] 10.00-12.00   sec  2.50 MBytes  10.5 Mbits/sec  0    383 KBytes
[ 7] 12.00-14.00   sec  1.62 MBytes  6.82 Mbits/sec  0    383 KBytes
[ 7] 14.00-16.00   sec  2.38 MBytes  9.96 Mbits/sec  0    383 KBytes
[ 7] 16.00-18.00   sec  2.38 MBytes  9.96 Mbits/sec  0    383 KBytes
[ 7] 18.00-20.00   sec  2.50 MBytes  10.5 Mbits/sec  0    383 KBytes
[ 7] 20.00-22.00   sec  2.38 MBytes  9.96 Mbits/sec  0    383 KBytes
[ 7] 22.00-24.00   sec  2.50 MBytes  10.5 Mbits/sec  0    383 KBytes
[ 7] 24.00-26.00   sec  1.50 MBytes  6.30 Mbits/sec  0    383 KBytes
[ 7] 26.00-28.00   sec  2.50 MBytes  10.5 Mbits/sec  0    383 KBytes
[ 7] 28.00-30.00   sec  2.38 MBytes  9.96 Mbits/sec  0    383 KBytes
--
[ ID] Interval      Transfer    Bitrate    Retr  sender receiver
[ 7] 0.00-30.00    sec  36.0 MBytes  10.1 Mbits/sec  0
[ 7] 0.00-30.13   sec  34.2 MBytes  9.54 Mbits/sec
```



C. time=30sec interval=5

```
root@vaishnavi:/home/vaishnavi# iperf3 -c 10.0.0.2 -t 30 -i 5
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 47344 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate    Retr  Cwnd
[ 7]  0.00-5.01    sec  7.75 MBytes  13.0 Mbits/sec  0    383 KBytes
[ 7]  5.01-10.01   sec  5.62 MBytes  9.44 Mbits/sec  0    383 KBytes
[ 7] 10.01-15.01   sec  5.75 MBytes  9.65 Mbits/sec  0    383 KBytes
[ 7] 15.01-20.01   sec  5.62 MBytes  9.44 Mbits/sec  0    383 KBytes
[ 7] 20.01-25.01   sec  5.62 MBytes  9.44 Mbits/sec  0    383 KBytes
[ 7] 25.01-30.01   sec  5.62 MBytes  9.44 Mbits/sec  0    383 KBytes
-----
[ ID] Interval      Transfer    Bitrate    Retr  sender
[ 7]  0.00-30.01   sec  36.0 MBytes 10.1 Mbits/sec  0
[ 7]  0.00-30.13   sec  34.4 MBytes 9.57 Mbits/sec  0    receiver
```



Topo3 :code

```
#!/usr/bin/python3

from mininet.topo import Topo
from mininet.net import Mininet
from mininet.link import TCLink
from mininet.node import OVSSwitch
from mininet.cli import CLI
from mininet.log import setLogLevel

class Topology3(Topo):
    """
    Topology-3:
        5Mbps      2Mbps      4Mbps
    h2 ---- s1 ----- s2 ---- h1

        |
        10Mbps
        |
        h3
    """

    def build(self):
```

```

h1 = self.addHost('h1') # Host1
h2 = self.addHost('h2') # Host2
h3 = self.addHost('h3') # Host3

s1 = self.addSwitch('s1', failMode='standalone')
s2 = self.addSwitch('s2', failMode='standalone')

self.addLink(h2, s1, cls=TCLink, bw=5) # 5 Mbps
self.addLink(s1, s2, cls=TCLink, bw=2) # 2 Mbps bottleneck
self.addLink(s2, h1, cls=TCLink, bw=4) # 4 Mbps
self.addLink(h3, s1, cls=TCLink, bw=10) # 10 Mbps

if __name__ == "__main__":
    setLogLevel('info')
    topo = Topology3()
    net = Mininet(
        topo=topo,
        controller=None, # no controller needed
        switch=OVSSwitch,
        link=TCLink,
        autoSetMacs=True,
        autoStaticArp=True
    )
    net.start()
    for sw in net.switches:
        sw.cmd('ovs-ofctl del-flows {sw.name}')
        sw.cmd('ovs-ofctl add-flow {sw.name} "priority=0,actions=NORMAL"')
    print("\n*** Running pingall")
    net.pingAll()
    print("\n*** Entering Mininet CLI")
    CLI(net)
    net.stop()

```

ii.

h2 traceroute h1

traceroute to 10.0.0.1 (10.0.0.1), 30 hops max, 60 byte packets

```

1  10.0.0.1 (10.0.0.1) 10.132 ms * *

```

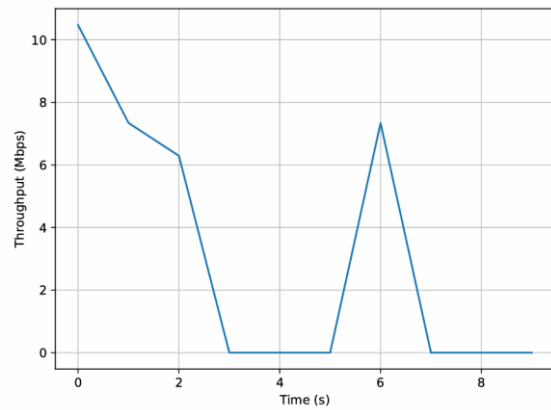
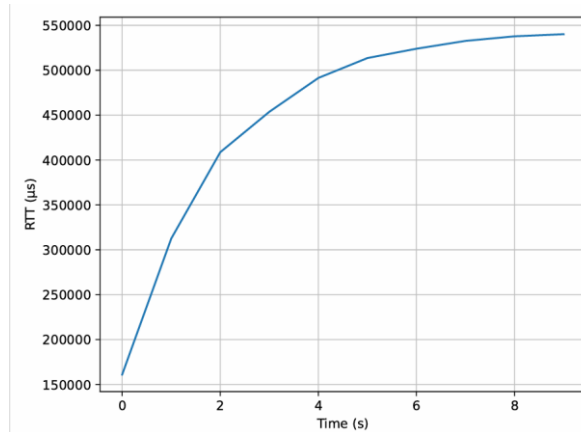
iii. throughput calculation

a. time =10sec interval=1

```

root@vaishnavi:/home/vaishnavi# iperf3 -c 10.0.0.2 -t 10 -i 1
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 40492 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate    Retr  Cwnd
[ 7] 0.00-1.00    sec  1.25 MBytes  10.5 Mbits/sec  0    256 KBytes
[ 7] 1.00-2.00    sec  896 KBytes  7.34 Mbits/sec  0    383 KBytes
[ 7] 2.00-3.00    sec  768 KBytes  6.29 Mbits/sec  0    383 KBytes
[ 7] 3.00-4.00    sec  0.00 Bytes  0.00 bits/sec  0    383 KBytes
[ 7] 4.00-5.00    sec  0.00 Bytes  0.00 bits/sec  0    383 KBytes
[ 7] 5.00-6.00    sec  0.00 Bytes  0.00 bits/sec  0    383 KBytes
[ 7] 6.00-7.00    sec  896 KBytes  7.34 Mbits/sec  0    383 KBytes
[ 7] 7.00-8.00    sec  0.00 Bytes  0.00 bits/sec  0    383 KBytes
[ 7] 8.00-9.00    sec  0.00 Bytes  0.00 bits/sec  0    383 KBytes
[ 7] 9.00-10.01   sec  0.00 Bytes  0.00 bits/sec  0    383 KBytes
-----
[ ID] Interval      Transfer    Bitrate    Retr
[ 7] 0.00-10.01   sec  3.75 MBytes  3.14 Mbits/sec  0
[ 7] 0.00-10.57   sec  2.38 MBytes  1.89 Mbits/sec
sender
receiver

```

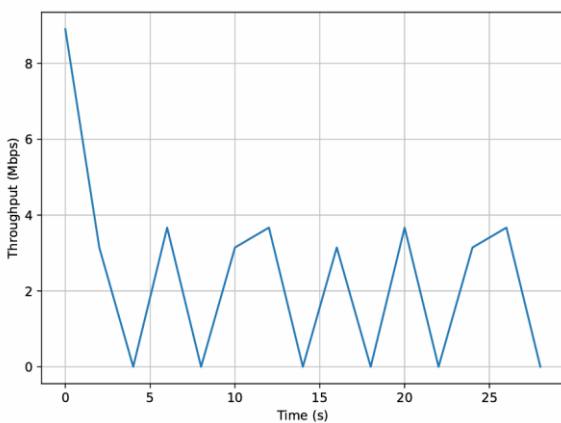
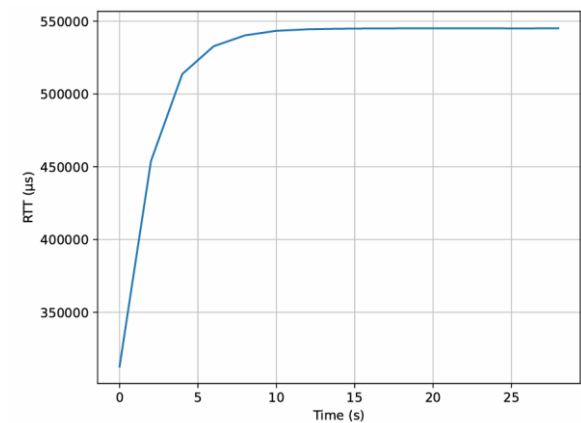


b. Time=30sec interval=2

```

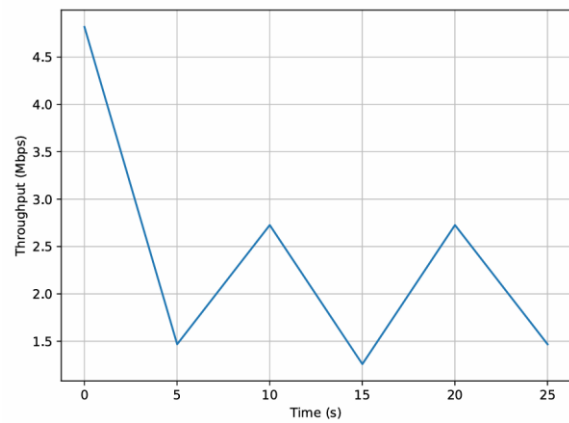
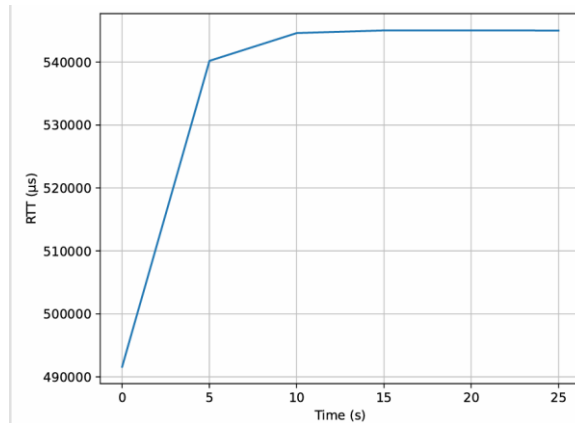
[ 7] local 10.0.0.1 port 52020 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate    Retr  Cwnd
[ 7] 0.00-2.00    sec  2.42 MBytes  8.90 Mbits/sec  0    383 KBytes
[ 7] 2.00-4.00    sec  768 KBytes  3.15 Mbits/sec  0    383 KBytes
[ 7] 4.00-6.00    sec  0.00 Bytes  0.00 bits/sec  0    383 KBytes
[ 7] 6.00-8.00    sec  896 KBytes  3.67 Mbits/sec  0    383 KBytes
[ 7] 8.00-10.00   sec  0.00 Bytes  0.00 bits/sec  0    383 KBytes
[ 7] 10.00-12.00   sec  768 KBytes  3.15 Mbits/sec  0    383 KBytes
[ 7] 12.00-14.00   sec  896 KBytes  3.67 Mbits/sec  0    383 KBytes
[ 7] 14.00-16.00   sec  0.00 Bytes  0.00 bits/sec  0    383 KBytes
[ 7] 16.00-18.00   sec  768 KBytes  3.15 Mbits/sec  0    383 KBytes
[ 7] 18.00-20.00   sec  0.00 Bytes  0.00 bits/sec  0    383 KBytes
[ 7] 20.00-22.00   sec  896 KBytes  3.67 Mbits/sec  0    383 KBytes
[ 7] 22.00-24.00   sec  0.00 Bytes  0.00 bits/sec  0    383 KBytes
[ 7] 24.00-26.00   sec  768 KBytes  3.15 Mbits/sec  0    383 KBytes
[ 7] 26.00-28.00   sec  896 KBytes  3.67 Mbits/sec  0    383 KBytes
[ 7] 28.00-30.00   sec  0.00 Bytes  0.00 bits/sec  0    383 KBytes
-----
[ ID] Interval      Transfer    Bitrate    Retr
[ 7] 0.00-30.00   sec  8.62 MBytes  2.41 Mbits/sec  0
[ 7] 0.00-30.74   sec  7.00 MBytes  1.91 Mbits/sec
sender
receiver

```



c. time=30sec interval=5

```
iperf Done.
root@vaishnavi:/home/vaishnavi# iperf3 -c 10.0.0.2 -t 30 -i 5
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 38398 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer      Bitrate      Retr  Cwnd
[ 7]  0.00-5.01    sec  2.88 MBytes  4.82 Mbits/sec  0    383 KBytes
[ 7]  5.01-10.01   sec  896 KBytes  1.47 Mbits/sec  0    383 KBytes
[ 7] 10.01-15.01   sec  1.62 MBytes  2.73 Mbits/sec  0    383 KBytes
[ 7] 15.01-20.01   sec  768 KBytes  1.26 Mbits/sec  0    383 KBytes
[ 7] 20.01-25.01   sec  1.62 MBytes  2.73 Mbits/sec  0    383 KBytes
[ 7] 25.01-30.01   sec  896 KBytes  1.47 Mbits/sec  0    383 KBytes
-----
[ ID] Interval      Transfer      Bitrate      Retr  sender receiver
[ 7]  0.00-30.01   sec  8.62 MBytes  2.41 Mbits/sec  0
[ 7]  0.00-30.74   sec  7.00 MBytes  1.91 Mbits/sec  0
```



Q2

Code

```
#!/usr/bin/python3

from mininet.topo import Topo
from mininet.net import Mininet
from mininet.link import TCLink
from mininet.node import OVSSwitch
from mininet.cli import CLI
from mininet.log import setLogLevel

class Q2Topo(Topo):
    """
    Q2 Topology:
    h1 --(5 Mbps)-- s1 --(10 Mbps)-- s2 --(2 Mbps)-- h2
    """

    def build(self):
        h1 = self.addHost('h1')
        h2 = self.addHost('h2')

        # standalone so it can work without controller binary
        s1 = self.addSwitch('s1', failMode='standalone')
```

```

s2 = self.addSwitch('s2', failMode='standalone')

self.addLink(h1, s1, cls=TCLink, bw=5)

self.addLink(s1, s2, cls=TCLink, bw=10)

self.addLink(s2, h2, cls=TCLink, bw=2) # bottleneck

if __name__ == "__main__":

    setLogLevel('info')

    topo = Q2Topo()

    net = Mininet(

        topo=topo,

        controller=None,

        switch=OVSSwitch,

        link=TCLink,

        autoSetMacs=True,

        autoStaticArp=True

    )

    net.start()

    for sw in net.switches:

        sw.cmd(f'ovs-ofctl del-flows {sw.name}')

        sw.cmd(f'ovs-ofctl add-flow {sw.name} "priority=0,actions=NORMAL"')

    print("\n*** Running pingall")

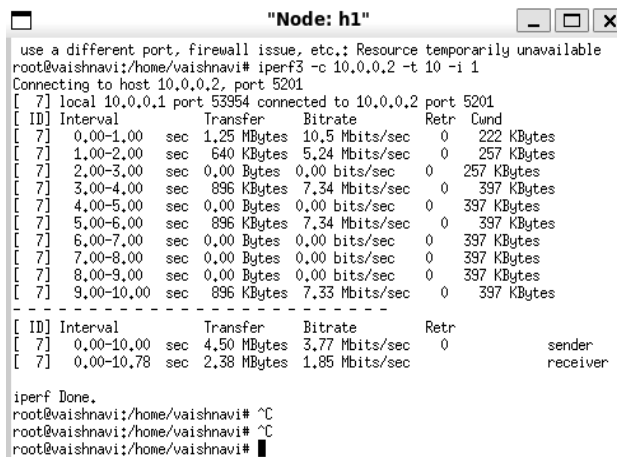
    net.pingAll()

    print("\n*** Entering Mininet CLI")

    CLI(net)

    net.stop()

```



```

Node: h1
use a different port, firewall issue, etc.: Resource temporarily unavailable
root@vaishnavi:/home/vaishnavi# iperf3 -c 10.0.0.2 -t 10 -i 1
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 53954 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate    Retr  Cwnd
[ 7] 0.00-1.00    sec  1.25 MBytes  10.5 Mbits/sec  0    222 KBytes
[ 7] 1.00-2.00    sec  640 KBytes  5.24 Mbits/sec  0    257 KBytes
[ 7] 2.00-3.00    sec  0.00 Bytes  0.00 bits/sec  0    257 KBytes
[ 7] 3.00-4.00    sec  896 KBytes  7.34 Mbits/sec  0    397 KBytes
[ 7] 4.00-5.00    sec  0.00 Bytes  0.00 bits/sec  0    397 KBytes
[ 7] 5.00-6.00    sec  896 KBytes  7.34 Mbits/sec  0    397 KBytes
[ 7] 6.00-7.00    sec  0.00 Bytes  0.00 bits/sec  0    397 KBytes
[ 7] 7.00-8.00    sec  0.00 Bytes  0.00 bits/sec  0    397 KBytes
[ 7] 8.00-9.00    sec  0.00 Bytes  0.00 bits/sec  0    397 KBytes
[ 7] 9.00-10.00   sec  896 KBytes  7.33 Mbits/sec  0    397 KBytes
-----
[ ID] Interval      Transfer    Bitrate    Retr
[ 7] 0.00-10.00   sec  4.50 MBytes  3.77 Mbits/sec  0
[ 7] 0.00-10.78   sec  2.38 MBytes  1.85 Mbits/sec
iperf Done.
root@vaishnavi:/home/vaishnavi# ^C
root@vaishnavi:/home/vaishnavi# ^C
root@vaishnavi:/home/vaishnavi#

```

3.

CODE

```
#!/usr/bin/python3

from mininet.topo import Topo
from mininet.net import Mininet
from mininet.link import TCLink
from mininet.node import OVSSwitch
from mininet.cli import CLI
from mininet.log import setLogLevel

class Q3CustomTopo(Topo):

    def build(self):

        h1 = self.addHost('h1') # Host-1
        h2 = self.addHost('h2') # Host-2
        h3 = self.addHost('h3') # Host-3
        h4 = self.addHost('h4') # Host-4

        s1 = self.addSwitch('s1', failMode='standalone')
        s2 = self.addSwitch('s2', failMode='standalone')

        self.addLink(h1, s1, cls=TCLink, bw=10)
        self.addLink(h3, s1, cls=TCLink, bw=5)
        self.addLink(s1, s2, cls=TCLink, bw=1) # BOTTLENECK
        self.addLink(s2, h2, cls=TCLink, bw=9)
        self.addLink(h4, s2, cls=TCLink, bw=3)

if __name__ == "__main__":

    setLogLevel('info')

    topo = Q3CustomTopo()

    net = Mininet(

        topo=topo,

        controller=None,

        switch=OVSSwitch,

        link=TCLink,

        autoSetMacs=True,

        autoStaticArp=True

    )
```

```
net.start()
```

```
for sw in net.switches:
```

```
    sw.cmd('ovs-ofctl del-flows {sw.name}')
```

```
sw.cmd('ovs-ofctl add-flow {sw.name} "priority=0,actions=NORMAL"')
```

```
print("\n*** Connectivity Test")
```

```
net.pingAll()
```

```
print("\n*** Entering Mininet CLI")
```

```
CLI(net)
```

```
net.stop()
```

ii.

```
mininet> h1 traceroute h2
```

```
traceroute to 10.0.0.2 (10.0.0.2), 30 hops max, 60 byte packets
```

```
1 10.0.0.2 (10.0.0.2) 0.340 ms 0.232 ms 0.054 ms
```

iii. throughput calculation

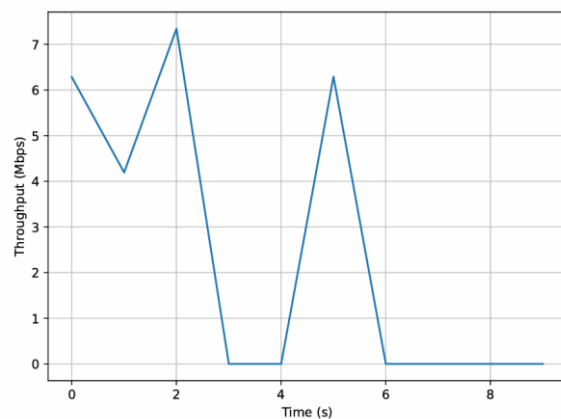
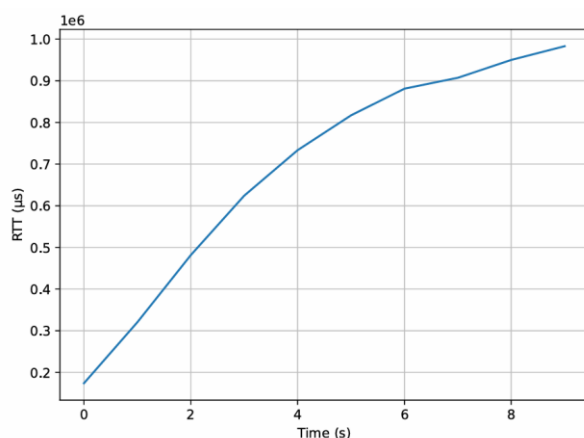
a. Time=10sec interval=1

```
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 60726 connected to 10.0.0.2 port 5201
```

ID	Interval	Transfer	Bitrate	Retr	Cwnd
[7]	0.00-1.00	sec 768 KBytes	6.28 Mbits/sec	0	136 KBytes
[7]	1.00-2.00	sec 512 KBytes	4.19 Mbits/sec	0	256 KBytes
[7]	2.00-3.00	sec 896 KBytes	7.34 Mbits/sec	0	383 KBytes
[7]	3.00-4.00	sec 0.00 Bytes	0.00 bits/sec	0	383 KBytes
[7]	4.00-5.00	sec 0.00 Bytes	0.00 bits/sec	0	383 KBytes
[7]	5.00-6.00	sec 768 KBytes	6.29 Mbits/sec	0	383 KBytes
[7]	6.00-7.00	sec 0.00 Bytes	0.00 bits/sec	0	383 KBytes
[7]	7.00-8.00	sec 0.00 Bytes	0.00 bits/sec	0	383 KBytes
[7]	8.00-9.00	sec 0.00 Bytes	0.00 bits/sec	0	383 KBytes
[7]	9.00-10.00	sec 0.00 Bytes	0.00 bits/sec	0	383 KBytes

ID	Interval	Transfer	Bitrate	Retr	sender	receiver
[7]	0.00-10.00	sec 2.88 MBytes	2.41 Mbits/sec	0		
[7]	0.00-11.33	sec 1.25 MBytes	926 Kbits/sec			

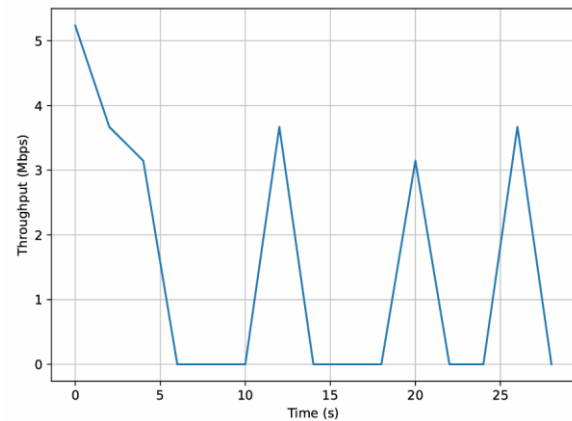
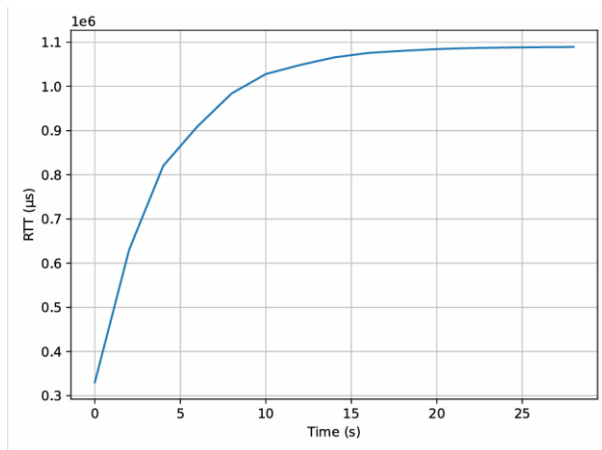
```
iperf Done.
```



b. time=30sec interval =2

ID	Interval	Transfer	Bitrate	Retr	Cwnd
7]	0,00-2,00	sec 1,25 MBytes	5,24 Mbits/sec	0	256 KBytes
7]	2,00-4,00	sec 896 KBytes	3,67 Mbits/sec	0	383 KBytes
7]	4,00-6,00	sec 768 KBytes	3,15 Mbits/sec	0	383 KBytes
7]	6,00-8,00	sec 0,00 Bytes	0,00 bits/sec	0	383 KBytes
7]	8,00-10,00	sec 0,00 Bytes	0,00 bits/sec	0	383 KBytes
7]	10,00-12,00	sec 0,00 Bytes	0,00 bits/sec	0	383 KBytes
7]	12,00-14,00	sec 896 KBytes	3,67 Mbits/sec	0	383 KBytes
7]	14,00-16,00	sec 0,00 Bytes	0,00 bits/sec	0	383 KBytes
7]	16,00-18,00	sec 0,00 Bytes	0,00 bits/sec	0	383 KBytes
7]	18,00-20,00	sec 0,00 Bytes	0,00 bits/sec	0	383 KBytes
7]	20,00-22,00	sec 768 KBytes	3,15 Mbits/sec	0	383 KBytes
7]	22,00-24,02	sec 0,00 Bytes	0,00 bits/sec	0	383 KBytes
7]	24,02-25,00	sec 0,00 Bytes	0,00 bits/sec	0	383 KBytes
7]	25,00-28,00	sec 896 KBytes	3,67 Mbits/sec	0	383 KBytes
7]	28,00-30,00	sec 0,00 Bytes	0,00 bits/sec	0	383 KBytes

ID	Interval	Transfer	Bitrate	Retr	
7]	0,00-30,00	sec 5,38 MBytes	1,50 Mbits/sec	0	sender
7]	0,00-31,50	sec 3,50 MBytes	932 Kbits/sec		receiver



c. time=30sec interval=5

```

root@vaishnavi:/home/vaishnavi# iperf3 -c 10.0.0.2 -t 30 -i 5
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 47900 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate    Retr    Cwnd
[ 7]  0,00-5,01    sec 2,12 MBytes  3,56 Mbits/sec  0    383 KBytes
[ 7]  5,01-10,01   sec 768 KBytes  1,26 Mbits/sec  0    383 KBytes
[ 7] 10,01-15,01   sec 896 KBytes  1,47 Mbits/sec  0    383 KBytes
[ 7] 15,01-20,01   sec 0,00 Bytes  0,00 bits/sec  0    383 KBytes
[ 7] 20,01-25,01   sec 768 KBytes  1,26 Mbits/sec  0    383 KBytes
[ 7] 25,01-30,01   sec 896 KBytes  1,47 Mbits/sec  0    383 KBytes

[ ID] Interval      Transfer    Bitrate    Retr
[ 7]  0,00-30,01   sec 5,38 MBytes  1,50 Mbits/sec  0
[ 7]  0,00-31,50   sec 3,50 MBytes  932 Kbits/sec

```

